

Table of Contents

1. Problem Formulation.....	1
2. DAG Orchestration in Airflow	1
3. Stagewise Details	2
4. Challenges	4

1. Problem Formulation

- Business Problem: Predict customer churn for a subscription-based company.
- Objective: Build an automated, end-to-end ML pipeline to ingest, validate, transform, and model customer data for churn prediction.
- Data Sources:
 - web_events.csv (customer online interactions)
 - subscriptions.json (subscription and transaction data)
- Outputs:
 - Clean & validated dataset for analysis
 - Transformed dataset with engineered features stored in Postgres
 - Trained ML model (Random Forest)
 - Deployed REST API for churn prediction
- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score

2. DAG Orchestration in Airflow

Airflow is used for end to end pipeline orchestration.

The DAG – Directed Acyclic Graph (churn_pipeline_astro_v5.py) defines the pipeline stages as tasks:

1. make_data_dirs → Ensures required folders exist
2. seed_raw_from_include → Loads raw data into working directory
3. ingest → Runs ingest.py to collect data into raw zone
4. prepare → Cleans data with prepare.py
5. validate → Validates prepared data (validate.py)
6. transform → Feature transformations (transform.py). Loads final transformed data into Postgres
7. build_feature_store → Creates feature store (build_feature_store.py)
8. train_model → Trains ML model (train.py)
9. evaluate_model → Evaluates trained model (evaluate.py)
10. deploy_model → Deploys REST API with FastAPI (deploy.py)

The DAG is scheduled to run daily at 21:45 PM IST.

Grid View showing All the successful runs of all stages

localhost:8080/dags/churn_pipeline_astro_v5/runs

Search Dags Ctrl+K Trigger

churn_pipeline_astro_v5

Schedule: 45 21 *** Latest Run: 2025-08-24, 21:45:00 Next Run: 2025-08-25, 21:45:00 Owner: astro Tags: ml, etl, astro, churn Latest Dag Version: v2

Overview Runs Tasks Backfills Events Code Details

All States All Run Types

Run After	State	Run Type	Start Date	End Date	Duration	Dag Version(s)
2025-08-24, 21:45:00	success	scheduled	2025-08-24, 21:45:00	2025-08-24, 21:45:40	00:00:40	v1
2025-08-23, 21:45:00	success	scheduled	2025-08-24, 21:31:02	2025-08-24, 21:32:25	00:01:22	v1

make_data_dirs
seed_raw_from_include
ingest
prepare
validate
transform
build_feature_store
train_model
evaluate_model
deploy_model

localhost:8080/dags/churn_pipeline_astro_v5/runs/scheduled_2025-08-24T16:15:00+00:00/

Search Dags Ctrl+K Trigger

churn_pipeline_astro_v5 Dag Run: 2025-08-24, 21:45:00

2025-08-24, 21:45:00 success

Add a note Clear Run Mark Run as...

Logical Date	Run Type	Start	End	Duration	Dag Version(s)
2025-08-24, 21:45:00	scheduled	2025-08-24, 21:45:00	2025-08-24, 21:45:40	00:00:40	v1

Task Instances Audit Logs Code Details Asset Events

Task ID	Map Index	State	Start Date	End Date	Try Number
transform		success	2025-08-24, 21:45:14	2025-08-24, 21:45:17	1
validate		success	2025-08-24, 21:45:12	2025-08-24, 21:45:13	1
prepare		success	2025-08-24, 21:45:03	2025-08-24, 21:45:12	1
ingest		success	2025-08-24, 21:45:02	2025-08-24, 21:45:02	1
seed_raw_from_include		success	2025-08-24, 21:45:01	2025-08-24, 21:45:02	1

make_data_dirs
seed_raw_from_include
ingest
prepare
validate
transform
build_feature_store
train_model
evaluate_model
deploy_model

The screenshot displays the Airflow web interface for a DAG named 'churn_pipeline_astro_v5'. The interface is in dark mode. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, and Admin. The main content area shows the 'Task Instances' tab for a successful run on 2025-08-24 at 21:45:00. A table lists the task instances, all of which are in a 'success' state.

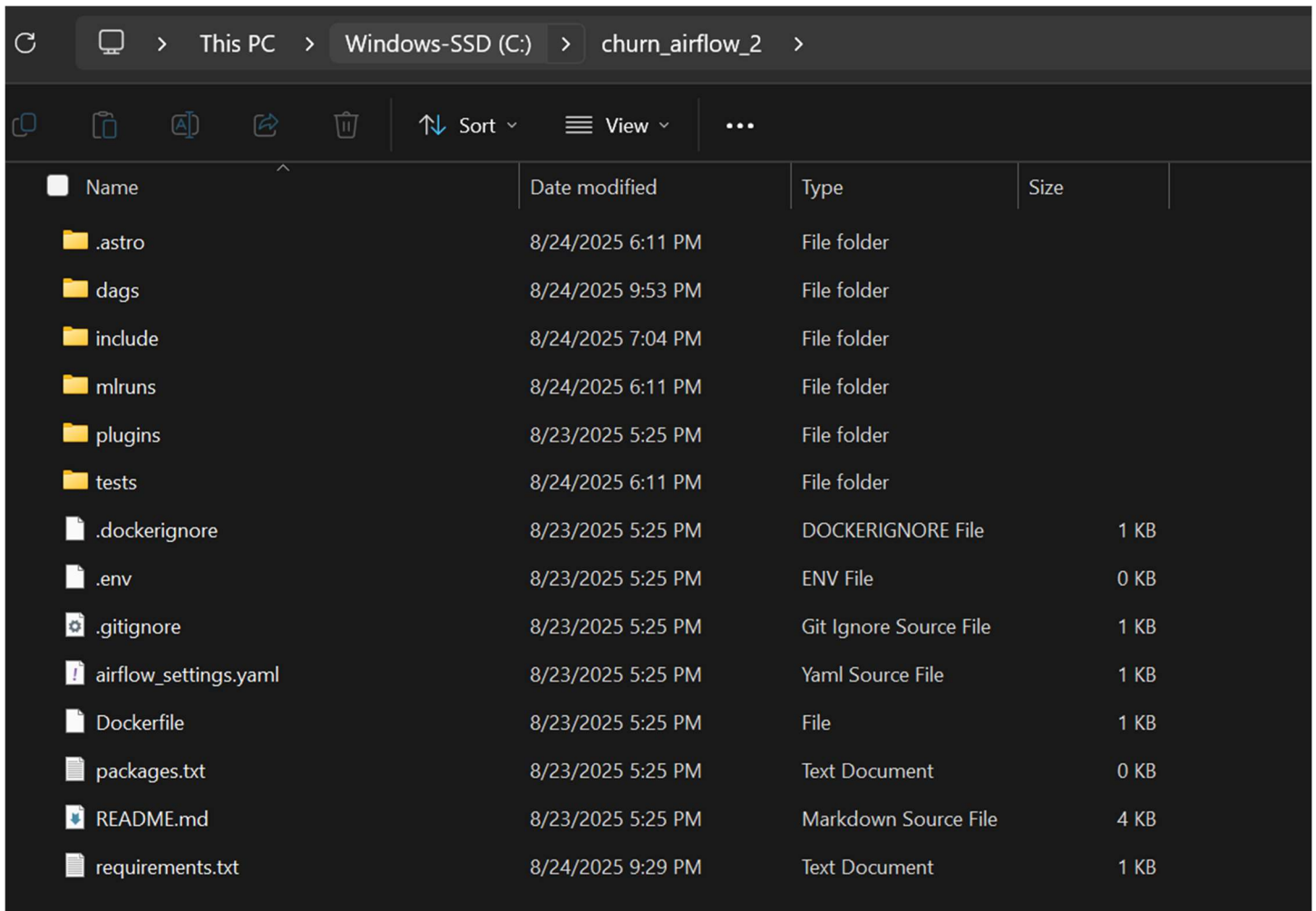
Task ID	Map Index	State	Start Date	End Date	Try Number
deploy_model		success	2025-08-24, 21:45:35	2025-08-24, 21:45:40	1
evaluate_model		success	2025-08-24, 21:45:28	2025-08-24, 21:45:34	1
train_model		success	2025-08-24, 21:45:19	2025-08-24, 21:45:27	1
build_feature_store		success	2025-08-24, 21:45:17	2025-08-24, 21:45:18	1

Graph View of DAG

The screenshot shows the 'Graph View' of the DAG 'churn_pipeline_astro_v5'. The DAG is visualized as a sequence of ten tasks connected by arrows, all in a 'success' state. The tasks are: make_data_dirs, seed_raw_from_include, ingest, prepare, validate, transform, build_feature_store, train_model, evaluate_model, and deploy_model. The right sidebar shows the DAG's metadata for a successful run on 2025-08-23 at 21:45:00.

Task ID	Task Name	Operator	State
make_data_dirs	make_data_dirs	PythonOperator	success
seed_raw_from_include	seed_raw_from_include	PythonOperator	success
ingest	ingest	BashOperator	success
prepare	prepare	BashOperator	success
validate	validate	BashOperator	success
transform	transform	BashOperator	success
build_feature_store	build_feature_store	BashOperator	success
train_model	train_model	BashOperator	success
evaluate_model	evaluate_model	BashOperator	success
deploy_model	deploy_model	BashOperator	success

The Airflow project structure using Astra



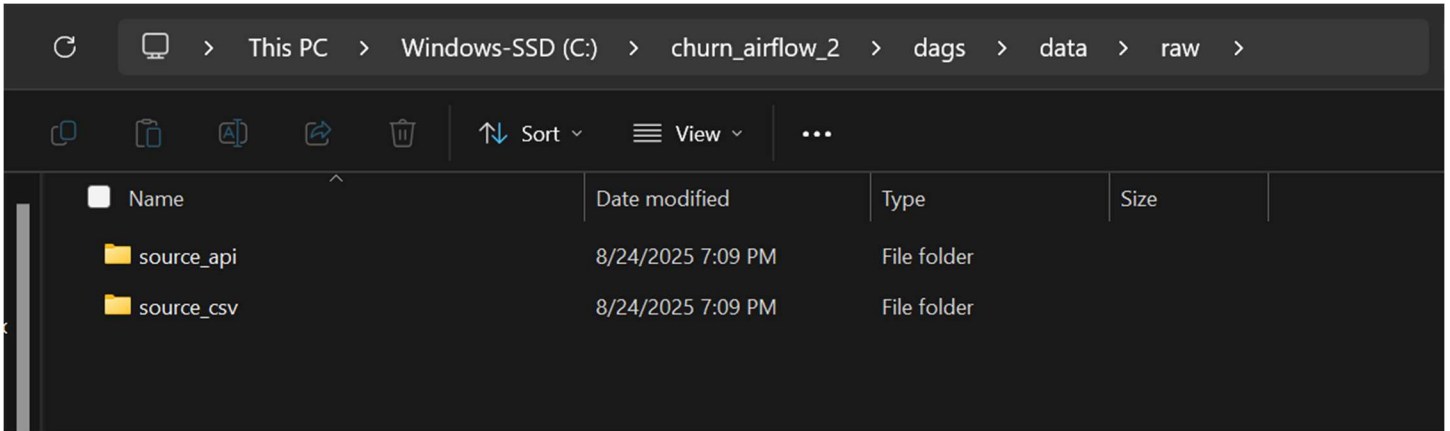
Name	Date modified	Type	Size
.astro	8/24/2025 6:11 PM	File folder	
dags	8/24/2025 9:53 PM	File folder	
include	8/24/2025 7:04 PM	File folder	
mlruns	8/24/2025 6:11 PM	File folder	
plugins	8/23/2025 5:25 PM	File folder	
tests	8/24/2025 6:11 PM	File folder	
.dockerignore	8/23/2025 5:25 PM	DOCKERIGNORE File	1 KB
.env	8/23/2025 5:25 PM	ENV File	0 KB
.gitignore	8/23/2025 5:25 PM	Git Ignore Source File	1 KB
airflow_settings.yaml	8/23/2025 5:25 PM	Yaml Source File	1 KB
Dockerfile	8/23/2025 5:25 PM	File	1 KB
packages.txt	8/23/2025 5:25 PM	Text Document	0 KB
README.md	8/23/2025 5:25 PM	Markdown Source File	4 KB
requirements.txt	8/24/2025 9:29 PM	Text Document	1 KB

3. Stage-wise Details

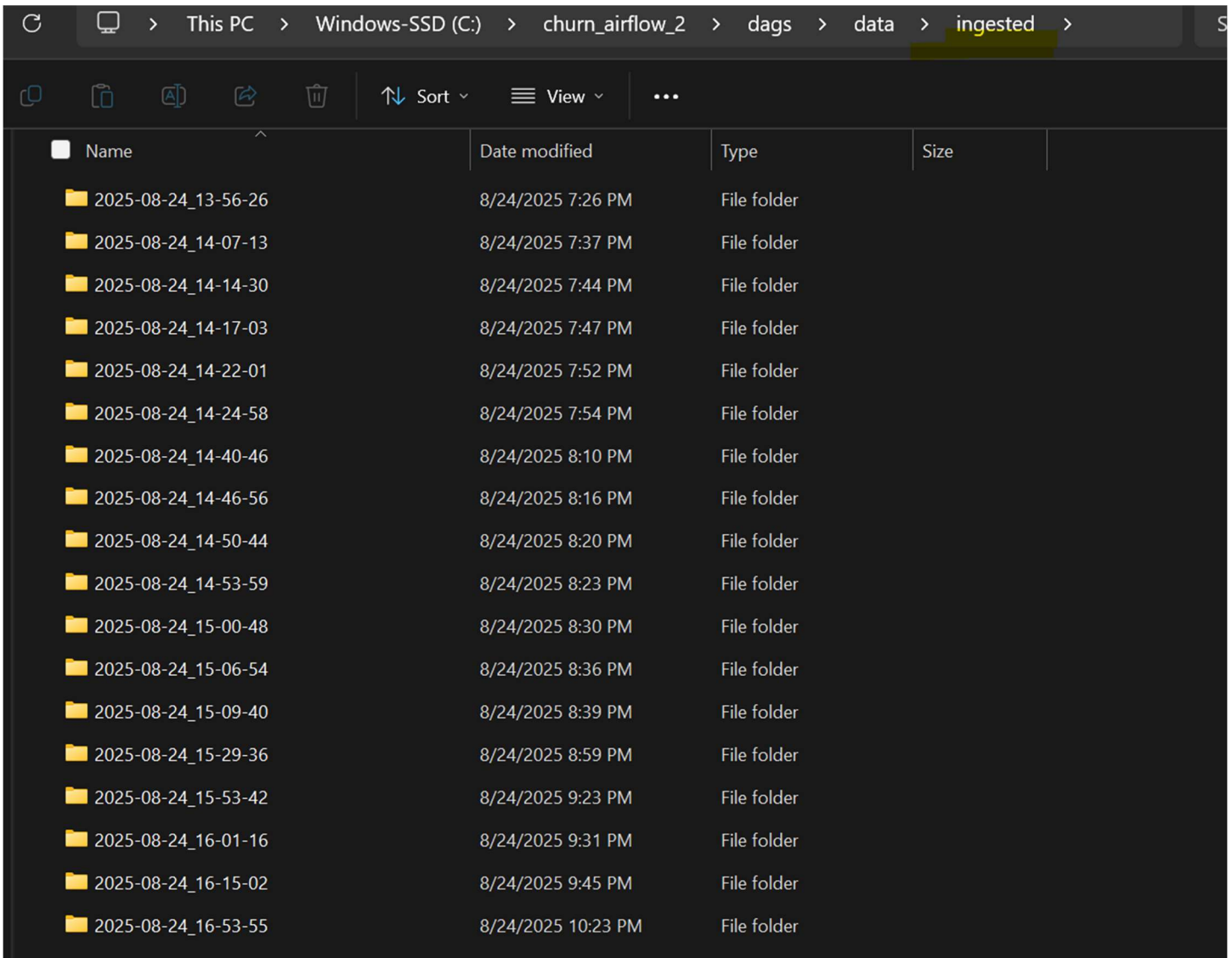
3.1 Data Ingestion (ingest.py)

- Purpose: Fetch raw customer data from multiple sources and store in the ingestion zone.
- Key Features:
 - Reads web_events.csv and subscriptions.json
 - Logs ingestion process to ingestion.log
 - Copies files into /usr/local/airflow/dags/data/ingested

Raw Data ingested in the local folder structure



Ingested data



churn_airflow_2 > dags > data > ingested > 2025-08-24_16-53-55				
📄 📁 🔍 🔗 🗑️ ↕️ Sort ☰ View ⋮				
Name	Date modified	Type	Size	
 subscriptions.json	8/24/2025 10:23 PM	JSON Source File	392 KB	
 web_events.csv	8/24/2025 10:23 PM	Microsoft Excel Com...	180 KB	

Ingestion Logs

```

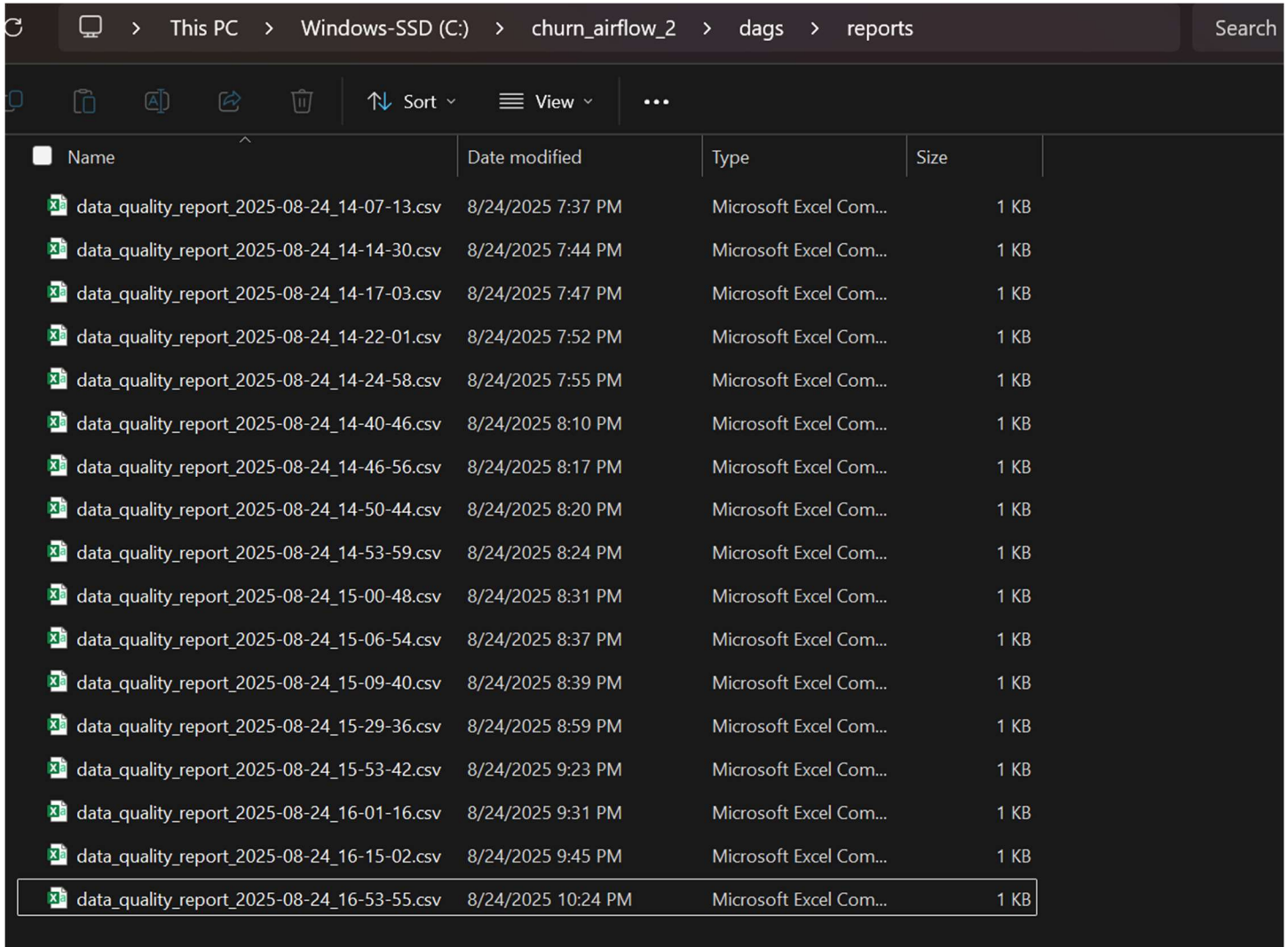
File Edit View
15-00-48/subscriptions.json
2025-08-24 15:00:48,294 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_15-00-48
2025-08-24 15:06:54,573 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-06-54/web_events.csv
2025-08-24 15:06:54,602 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-06-54/subscriptions.json
2025-08-24 15:06:54,603 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_15-06-54
2025-08-24 15:09:40,035 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-09-40/web_events.csv
2025-08-24 15:09:40,078 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-09-40/subscriptions.json
2025-08-24 15:09:40,079 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_15-09-40
2025-08-24 15:29:36,338 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-29-36/web_events.csv
2025-08-24 15:29:36,386 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-29-36/subscriptions.json
2025-08-24 15:29:36,386 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_15-29-36
2025-08-24 15:53:42,947 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-53-42/web_events.csv
2025-08-24 15:53:42,974 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_15-53-42/subscriptions.json
2025-08-24 15:53:42,974 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_15-53-42
2025-08-24 16:01:16,720 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-01-16/web_events.csv
2025-08-24 16:01:16,820 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-01-16/subscriptions.json
2025-08-24 16:01:16,821 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_16-01-16
2025-08-24 16:15:02,680 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-15-02/web_events.csv
2025-08-24 16:15:02,711 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-15-02/subscriptions.json
2025-08-24 16:15:02,712 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_16-15-02
2025-08-24 16:53:55,148 INFO Ingested /usr/local/airflow/dags/data/raw/source_csv/web_events.csv -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-53-55/web_events.csv
2025-08-24 16:53:55,177 INFO Ingested /usr/local/airflow/dags/data/raw/source_api/subscriptions.json -> /usr/local/airflow/dags/data/ingested/2025-08-24_16-53-55/subscriptions.json
2025-08-24 16:53:55,178 INFO Ingestion complete. Files copied to /usr/local/airflow/dags/data/ingested/2025-08-24_16-53-55

```

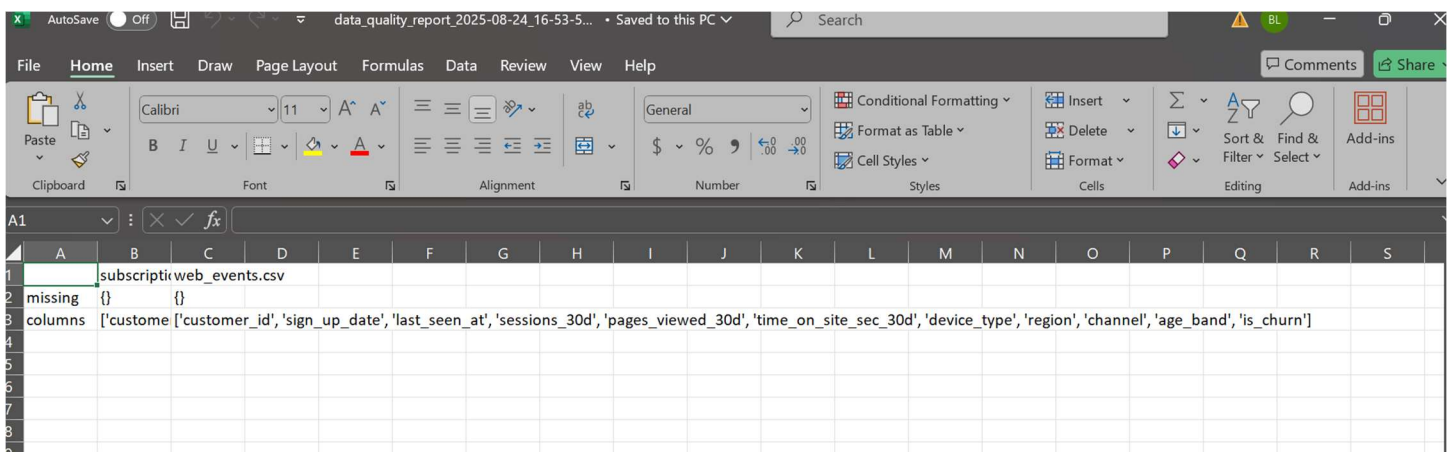
3.2 Data Validation (validate.py)

- Purpose: Ensure data quality before further processing.
- Checks Implemented:
 - Missing values
 - Duplicates
 - Schema validation (data types, ranges)
- Outputs:
 - Validation report (validation.log and CSV in /reports)

Validation Reports



Name	Date modified	Type	Size
data_quality_report_2025-08-24_14-07-13.csv	8/24/2025 7:37 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-14-30.csv	8/24/2025 7:44 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-17-03.csv	8/24/2025 7:47 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-22-01.csv	8/24/2025 7:52 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-24-58.csv	8/24/2025 7:55 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-40-46.csv	8/24/2025 8:10 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-46-56.csv	8/24/2025 8:17 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-50-44.csv	8/24/2025 8:20 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_14-53-59.csv	8/24/2025 8:24 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_15-00-48.csv	8/24/2025 8:31 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_15-06-54.csv	8/24/2025 8:37 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_15-09-40.csv	8/24/2025 8:39 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_15-29-36.csv	8/24/2025 8:59 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_15-53-42.csv	8/24/2025 9:23 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_16-01-16.csv	8/24/2025 9:31 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_16-15-02.csv	8/24/2025 9:45 PM	Microsoft Excel Com...	1 KB
data_quality_report_2025-08-24_16-53-55.csv	8/24/2025 10:24 PM	Microsoft Excel Com...	1 KB



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	subscription_events.csv																	
2	missing {}																	
3	columns ['customer_id', 'sign_up_date', 'last_seen_at', 'sessions_30d', 'pages_viewed_30d', 'time_on_site_sec_30d', 'device_type', 'region', 'channel', 'age_band', 'is_churn']																	
4																		
5																		
6																		
7																		
8																		
9																		

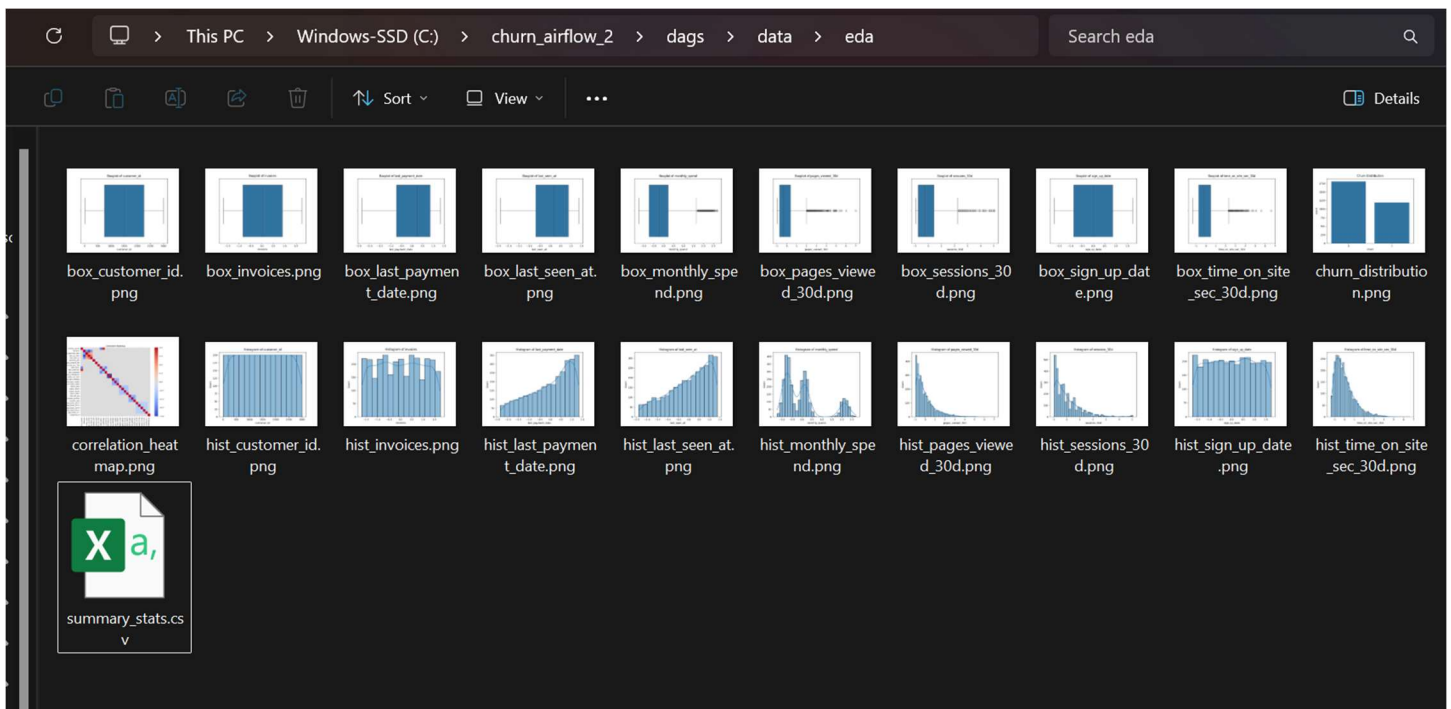
Validation Logs


```
2025-08-24 14:07:28,134 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-07-13
2025-08-24 14:07:28,269 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-07-13.csv
2025-08-24 14:14:42,056 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-14-30
2025-08-24 14:14:42,129 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-14-30.csv
2025-08-24 14:17:24,740 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-17-03
2025-08-24 14:17:24,824 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-17-03.csv
2025-08-24 14:22:12,786 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-22-01
2025-08-24 14:22:12,830 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-22-01.csv
2025-08-24 14:25:22,151 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-24-58
2025-08-24 14:25:22,264 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-24-58.csv
2025-08-24 14:40:59,288 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-40-46
2025-08-24 14:40:59,359 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-40-46.csv
2025-08-24 14:47:07,942 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-46-56
2025-08-24 14:47:08,027 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-46-56.csv
2025-08-24 14:50:55,097 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-50-44
2025-08-24 14:50:55,163 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-50-44.csv
2025-08-24 14:54:20,242 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_14-53-59
2025-08-24 14:54:20,337 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_14-53-59.csv
2025-08-24 15:01:03,769 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_15-00-48
2025-08-24 15:01:03,837 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_15-00-48.csv
2025-08-24 15:07:06,345 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_15-06-54
2025-08-24 15:07:06,461 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_15-06-54.csv
2025-08-24 15:09:50,418 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_15-09-40
2025-08-24 15:09:50,470 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_15-09-40.csv
2025-08-24 15:29:49,853 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_15-29-36
2025-08-24 15:29:49,904 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_15-29-36.csv
2025-08-24 15:53:56,142 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_15-53-42
2025-08-24 15:53:56,226 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_15-53-42.csv
2025-08-24 16:01:54,604 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_16-01-16
2025-08-24 16:01:54,682 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_16-01-16.csv
2025-08-24 16:15:13,641 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_16-15-02
2025-08-24 16:15:13,701 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_16-15-02.csv
2025-08-24 16:54:06,153 INFO Validating files in /usr/local/airflow/dags/data/ingested/2025-08-24_16-53-55
2025-08-24 16:54:06,212 INFO ✅ Data validation complete. Report saved at /usr/local/airflow/dags/reports/data_quality_report_2025-08-24_16-53-55.csv
```

3.3 Data Preparation (prepare.py)

- Purpose: Clean and preprocess data for ML.
- Steps:
 - Handle missing values (imputation/removal)
 - Encode categorical variables (LabelEncoder, One-hot encoding)
 - Scale numerical attributes (StandardScaler)
 - Generate EDA plots (saved in /data/eda/)
- Output: prepared.csv

Summary Stats and EDA charts

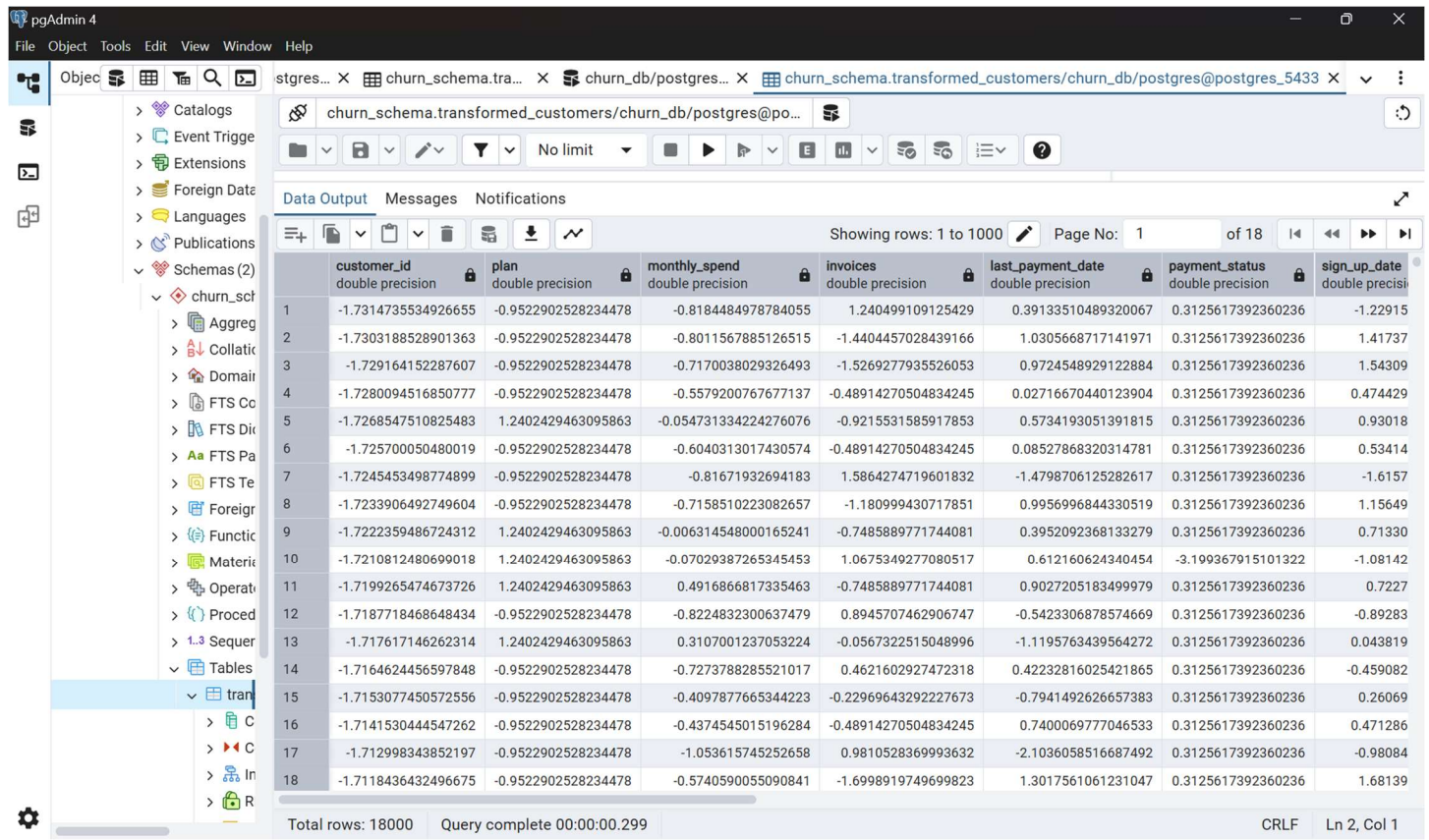


3.4 Data Transformation (transform.py)

- Purpose: Feature engineering and normalization.
- Steps:
 - Create derived features (e.g., customer tenure, activity frequency)
 - Standardize features
 - Save transformed dataset to /data/transformed/

Transformed Data

This PC > Windows-SSD (C:) > churn_airflow_2 > dags > data > transformed				
Name	Date modified	Type	Size	
sample_queries.sql	8/24/2025 10:24 PM	SQL Source File	1 KB	
schema.sql	8/24/2025 10:24 PM	SQL Source File	1 KB	
transformation_summary.md	8/24/2025 10:24 PM	Markdown Source File	1 KB	
transformed.csv	8/24/2025 10:24 PM	Microsoft Excel Com...	1,154 KB	

Transformed Data stored to Postgres database

pgAdmin 4

File Object Tools Edit View Window Help

churn_data.transformed_customers/churn_db/postgres@postgres_5433

Showing rows: 1 to 1000 Page No: 1 of 18

	customer_id double precision	plan double precision	monthly_spend double precision	invoices double precision	last_payment_date double precision	payment_status double precision	sign_up_date double precision
1	-1.7314735534926655	-0.9522902528234478	-0.8184484978784055	1.240499109125429	0.39133510489320067	0.3125617392360236	-1.22915
2	-1.7303188528901363	-0.9522902528234478	-0.8011567885126515	-1.4404457028439166	1.0305668717141971	0.3125617392360236	1.41737
3	-1.729164152287607	-0.9522902528234478	-0.7170038029326493	-1.5269277935526053	0.9724548929122884	0.3125617392360236	1.54309
4	-1.7280094516850777	-0.9522902528234478	-0.5579200767677137	-0.48914270504834245	0.02716670440123904	0.3125617392360236	0.474429
5	-1.7268547510825483	1.2402429463095863	-0.054731334224276076	-0.9215531585917853	0.5734193051391815	0.3125617392360236	0.93018
6	-1.725700050480019	-0.9522902528234478	-0.6040313017430574	-0.48914270504834245	0.08527868320314781	0.3125617392360236	0.53414
7	-1.7245453498774899	-0.9522902528234478	-0.81671932694183	1.5864274719601832	-1.4798706125282617	0.3125617392360236	-1.6157
8	-1.7233906492749604	-0.9522902528234478	-0.7158510223082657	-1.180999430717851	0.9956996844330519	0.3125617392360236	1.15649
9	-1.7222359486724312	1.2402429463095863	-0.006314548000165241	-0.7485889771744081	0.3952092368133279	0.3125617392360236	0.71330
10	-1.7210812480699018	1.2402429463095863	-0.07029387265345453	1.0675349277080517	0.612160624340454	-3.199367915101322	-1.08142
11	-1.7199265474673726	1.2402429463095863	0.4916866817335463	-0.7485889771744081	0.9027205183499979	0.3125617392360236	0.7227
12	-1.7187718468648434	-0.9522902528234478	-0.8224832300637479	0.8945707462906747	-0.5423306878574669	0.3125617392360236	-0.89283
13	-1.717617146262314	1.2402429463095863	0.3107001237053224	-0.0567322515048996	-1.1195763439564272	0.3125617392360236	0.043819
14	-1.7164624456597848	-0.9522902528234478	-0.7273788285521017	0.4621602927472318	0.42232816025421865	0.3125617392360236	-0.459082
15	-1.7153077450572556	-0.9522902528234478	-0.4097877665344223	-0.22969643292227673	-0.7941492626657383	0.3125617392360236	0.26069
16	-1.7141530444547262	-0.9522902528234478	-0.4374545015196284	-0.48914270504834245	0.7400069777046533	0.3125617392360236	0.471286
17	-1.712998343852197	-0.9522902528234478	-1.053615745252658	0.9810528369993632	-2.1036058516687492	0.3125617392360236	-0.98084
18	-1.7118436432496675	-0.9522902528234478	-0.5740590055090841	-1.6998919749699823	1.3017561061231047	0.3125617392360236	1.68139

Total rows: 18000 Query complete 00:00:00.299 CRLF Ln 2, Col 1

Models

Name	Date modified	Type	Size
random_forest_20250824_165415.pkl	8/24/2025 10:24 PM	PKL File	16,725 KB
logistic_regression_20250824_165415.pkl	8/24/2025 10:24 PM	PKL File	2 KB
features_20250824_165415.json	8/24/2025 10:24 PM	JSON Source File	1 KB
logistic_regression_20250824_161523.pkl	8/24/2025 9:45 PM	PKL File	2 KB
random_forest_20250824_161523.pkl	8/24/2025 9:45 PM	PKL File	16,725 KB
features_20250824_161523.json	8/24/2025 9:45 PM	JSON Source File	1 KB
random_forest_20250824_160206.pkl	8/24/2025 9:32 PM	PKL File	16,725 KB
logistic_regression_20250824_160206.pkl	8/24/2025 9:32 PM	PKL File	2 KB
features_20250824_160206.json	8/24/2025 9:32 PM	JSON Source File	1 KB
logistic_regression_20250824_155407.pkl	8/24/2025 9:24 PM	PKL File	2 KB
random_forest_20250824_155407.pkl	8/24/2025 9:24 PM	PKL File	16,725 KB
features_20250824_155407.json	8/24/2025 9:24 PM	JSON Source File	1 KB
logistic_regression_20250824_153124.pkl	8/24/2025 9:01 PM	PKL File	2 KB
random_forest_20250824_153124.pkl	8/24/2025 9:01 PM	PKL File	16,725 KB
features_20250824_153124.json	8/24/2025 9:01 PM	JSON Source File	1 KB
logistic_regression_20250824_153000.pkl	8/24/2025 9:00 PM	PKL File	2 KB
random_forest_20250824_153000.pkl	8/24/2025 9:00 PM	PKL File	16,725 KB
features_20250824_153000.json	8/24/2025 9:00 PM	JSON Source File	1 KB
random_forest_20250824_151002.pkl	8/24/2025 8:40 PM	PKL File	16,725 KB

3.5 Feature Store (build_feature_store.py, feature_store.py)

- Purpose: Maintain centralized repository of engineered features.
- Implementation:
 - Uses a custom LocalFeatureStore class
 - Stores metadata: feature name, version, description
 - Enables feature retrieval for training & inference
- Feature Store

File Explorer view of the `feature_store` directory:

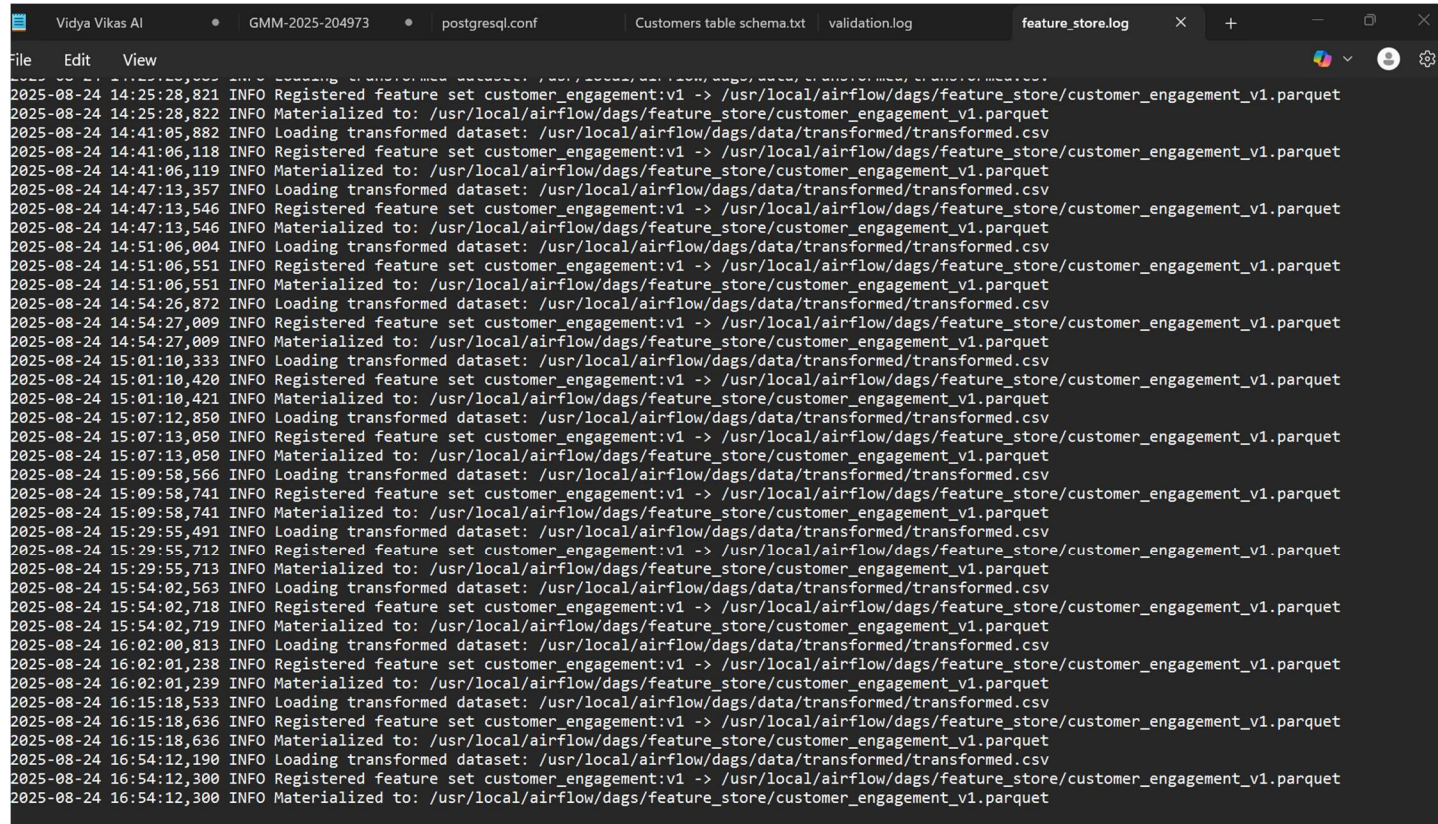
Name	Date modified	Type	Size
customer_engagement_v1.parquet	8/24/2025 10:24 PM	PARQUET File	188 KB
registry.yaml	8/24/2025 10:24 PM	Yaml Source File	2 KB

Code editor view of `registry.yaml`:

```

1 feature_sets:
2   customer_engagement:v1:
3     name: customer_engagement
4     version: v1
5     description: Customer engagement & value engineered features (v1).
6     source: dags/scripts/transform.py
7     primary_key: customer_id
8     features:
9       avg_monthly_spend_per_invoice:
10        dtype: float64
11        description: Monthly spend normalized by invoice count.
12      avg_monthly_spend_per_invoice_scaled:
13        dtype: float64
14        description: Feature `avg_monthly_spend_per_invoice_scaled`.
15      engagement_score:
16        dtype: float64
17        description: Composite engagement metric from sessions, pages and t
18        site.
19      engagement_score_scaled:
20        dtype: float64
21        description: Feature `engagement_score_scaled`.
22      tenure_proxy:
23        dtype: float64
24        description: 'Proxy for tenure: last_seen_at - sign_up_date.'
25      tenure_proxy_scaled:
26        dtype: float64
27        description: Feature `tenure_proxy_scaled`.
28      materialization_path: /usr/local/airflow/dags/feature_store/customer_en
29

```



The screenshot shows a terminal window with a dark background. The title bar at the top includes the name 'Vidya Vikas AI' and several open tabs: 'GMM-2025-204973', 'postgresql.conf', 'Customers table schema.txt', 'validation.log', and 'feature_store.log'. The main content area displays a log stream with timestamps and log messages. The messages are repetitive, showing a cycle of 'Registered feature set', 'Materialized to', and 'Loading transformed dataset' for a feature set named 'customer_engagement:v1'. The paths mentioned in the logs are under '/usr/local/airflow/dags/feature_store/' and '/usr/local/airflow/dags/data/transformed/'. The log entries are timestamped with dates in 2025-08-24 and times ranging from 14:25 to 16:54.

```
2025-08-24 14:25:28,821 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:25:28,822 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:41:05,882 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 14:41:06,118 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:41:06,119 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:47:13,357 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 14:47:13,546 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:47:13,546 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:51:06,004 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 14:51:06,551 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:51:06,551 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:54:26,872 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 14:54:27,009 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 14:54:27,009 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:01:10,333 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 15:01:10,420 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:01:10,421 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:07:12,850 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 15:07:13,050 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:07:13,050 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:09:58,566 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 15:09:58,741 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:09:58,741 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:29:55,491 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 15:29:55,712 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:29:55,713 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:54:02,563 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 15:54:02,718 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 15:54:02,719 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:02:00,813 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 16:02:01,238 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:02:01,239 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:15:18,533 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 16:15:18,636 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:15:18,636 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:54:12,190 INFO Loading transformed dataset: /usr/local/airflow/dags/data/transformed/transformed.csv
2025-08-24 16:54:12,300 INFO Registered feature set customer_engagement:v1 -> /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
2025-08-24 16:54:12,300 INFO Materialized to: /usr/local/airflow/dags/feature_store/customer_engagement_v1.parquet
```

3.6 Model Training (train.py)

- Purpose: Train ML model to predict churn.
- Steps:
 - Load transformed data
 - Split into training/test sets
 - Train Logistic Regression and Random Forest models
 - Select best model (based on F1-score)
 - Save model as model.pkl in /data/models/

Model Training Logs

```
File Edit View
2025-08-24 15:10:05,882 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_151002.pkl
2025-08-24 15:10:06,403 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_151002.pkl
2025-08-24 15:30:00,691 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 15:30:00,698 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_153000.json
2025-08-24 15:30:02,372 INFO Trained logistic_regression
2025-08-24 15:30:04,116 INFO Trained random_forest
2025-08-24 15:30:04,134 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_153000.pkl
2025-08-24 15:30:04,990 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_153000.pkl
2025-08-24 15:31:24,960 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 15:31:24,975 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_153124.json
2025-08-24 15:31:26,621 INFO Trained logistic_regression
2025-08-24 15:31:29,228 INFO Trained random_forest
2025-08-24 15:31:29,235 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_153124.pkl
2025-08-24 15:31:29,755 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_153124.pkl
2025-08-24 15:54:07,349 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 15:54:07,358 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_155407.json
2025-08-24 15:54:08,695 INFO Trained logistic_regression
2025-08-24 15:54:10,392 INFO Trained random_forest
2025-08-24 15:54:10,398 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_155407.pkl
2025-08-24 15:54:10,972 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_155407.pkl
2025-08-24 16:02:06,355 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 16:02:06,363 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_160206.json
2025-08-24 16:02:07,813 INFO Trained logistic_regression
2025-08-24 16:02:09,784 INFO Trained random_forest
2025-08-24 16:02:09,813 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_160206.pkl
2025-08-24 16:02:10,463 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_160206.pkl
2025-08-24 16:15:23,096 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 16:15:23,102 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_161523.json
2025-08-24 16:15:24,726 INFO Trained logistic_regression
2025-08-24 16:15:26,338 INFO Trained random_forest
2025-08-24 16:15:26,345 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_161523.pkl
2025-08-24 16:15:26,957 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_161523.pkl
2025-08-24 16:54:15,842 INFO Loaded /usr/local/airflow/dags/data/transformed/transformed.csv shape=(3000, 36)
2025-08-24 16:54:15,877 INFO Saved feature list -> /usr/local/airflow/dags/data/models/features_20250824_165415.json
2025-08-24 16:54:17,172 INFO Trained logistic_regression
2025-08-24 16:54:18,767 INFO Trained random_forest
2025-08-24 16:54:18,816 INFO Saved logistic_regression -> /usr/local/airflow/dags/data/models/logistic_regression_20250824_165415.pkl
2025-08-24 16:54:19,324 INFO Saved random_forest -> /usr/local/airflow/dags/data/models/random_forest_20250824_165415.pkl
```

3.7 Model Evaluation (evaluate.py)

- Purpose: Evaluate trained model performance.
- Metrics: Accuracy, Precision, Recall, F1-score
- Outputs:
 - Evaluation report (JSON/CSV)
 - Visuals (confusion matrix, ROC curve, feature importance)

Model Evaluation

This PC

Windows-SSD (C:)

churn_airflow_2

dags

data

evaluation

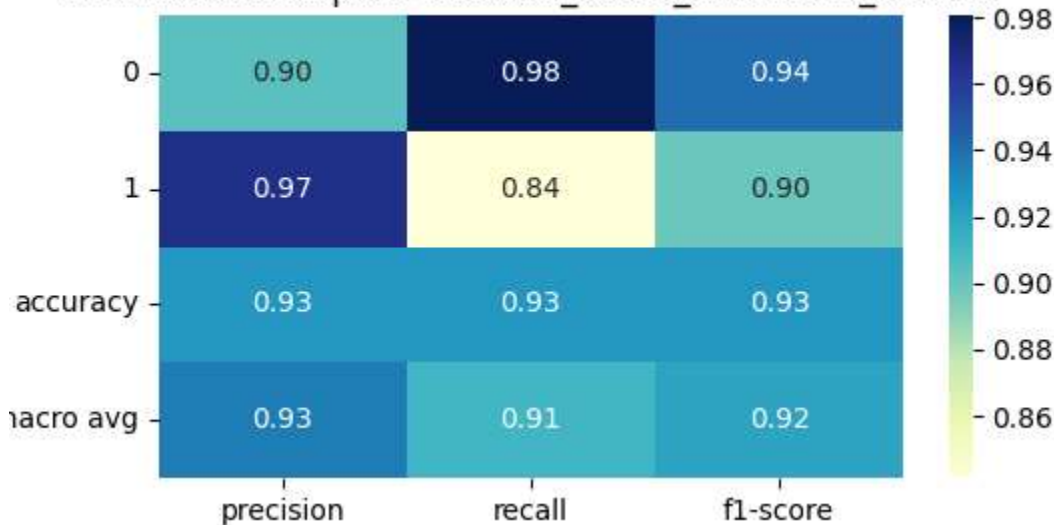
Search evaluation

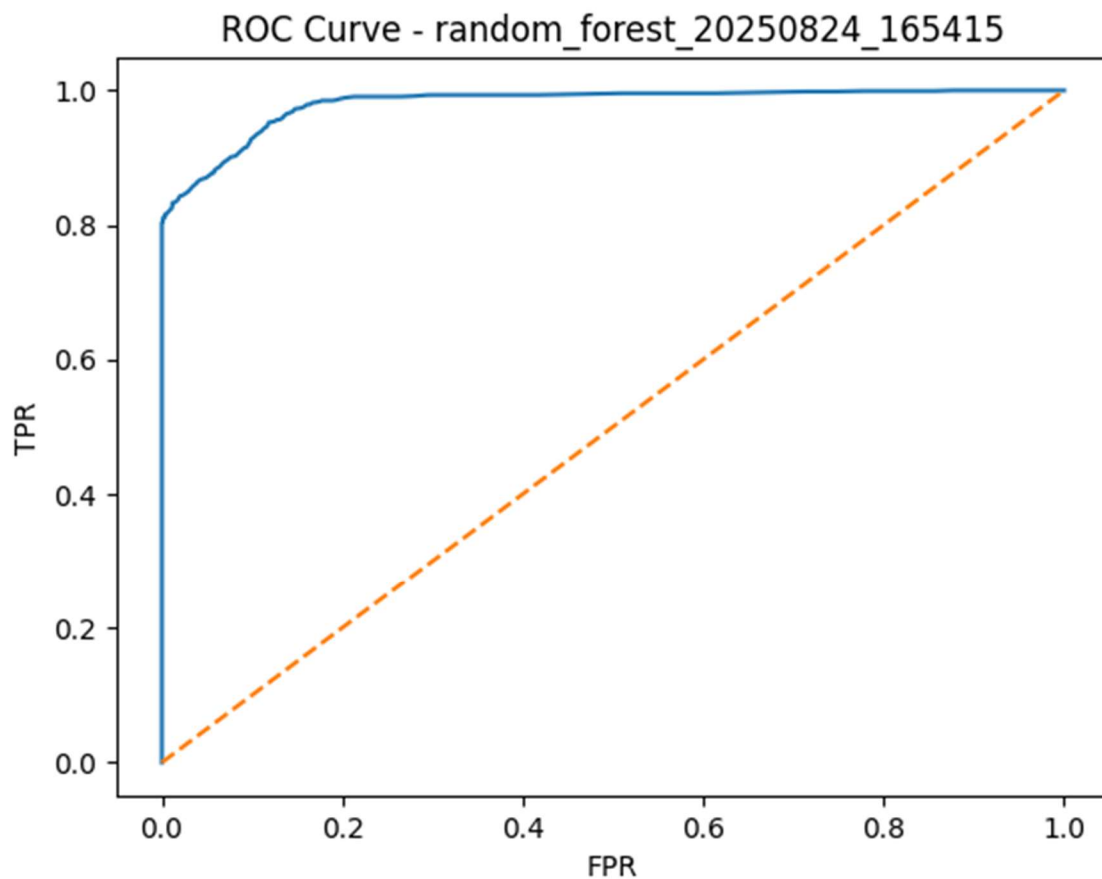
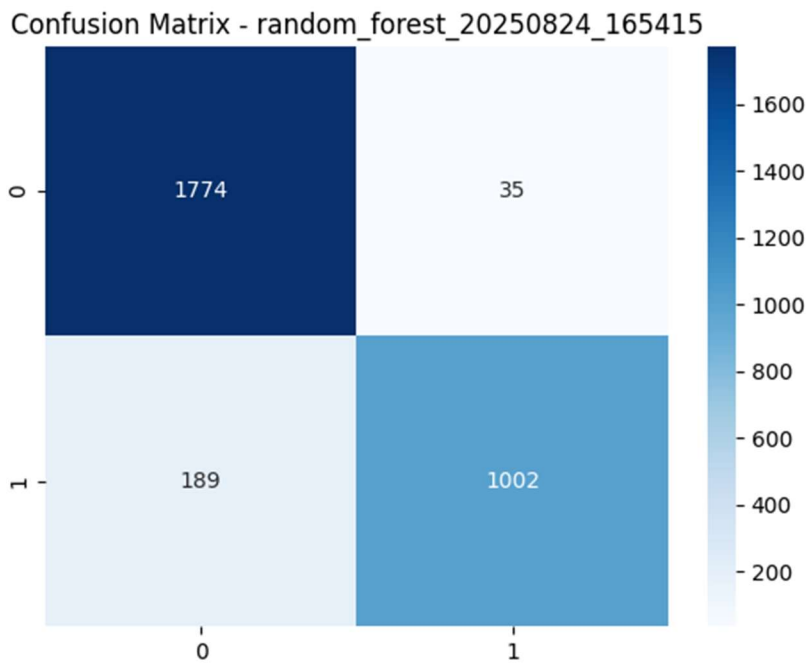
Sort

View

Name	Date	Type	Size	Tags
 random_forest_20250824_165415_report_20250824_165424.csv	8/24/2025 10:24 PM	Microsoft Excel Com...	1 KB	
 random_forest_20250824_165415_metrics_20250824_165424.json	8/24/2025 10:24 PM	JSON Source File	1 KB	
 random_forest_20250824_165415_report_20250824_165424.png	8/24/2025 10:24 PM	PNG File	29 KB	
 random_forest_20250824_165415_cm_20250824_165424.png	8/24/2025 10:24 PM	PNG File	22 KB	
 random_forest_20250824_165415_roc_20250824_165424.png	8/24/2025 10:24 PM	PNG File	25 KB	
 random_forest_20250824_161523_report_20250824_161532.csv	8/24/2025 9:45 PM	Microsoft Excel Com...	1 KB	
 random_forest_20250824_161523_metrics_20250824_161532.json	8/24/2025 9:45 PM	JSON Source File	1 KB	
 random_forest_20250824_161523_report_20250824_161532.png	8/24/2025 9:45 PM	PNG File	29 KB	
 random_forest_20250824_161523_cm_20250824_161532.png	8/24/2025 9:45 PM	PNG File	22 KB	
 random_forest_20250824_161523_roc_20250824_161532.png	8/24/2025 9:45 PM	PNG File	25 KB	
 random_forest_20250824_160206_report_20250824_160216.csv	8/24/2025 9:32 PM	Microsoft Excel Com...	1 KB	
 random_forest_20250824_160206_metrics_20250824_160216.json	8/24/2025 9:32 PM	JSON Source File	1 KB	
 random_forest_20250824_160206_report_20250824_160216.png	8/24/2025 9:32 PM	PNG File	29 KB	
 random_forest_20250824_160206_cm_20250824_160216.png	8/24/2025 9:32 PM	PNG File	22 KB	
 random_forest_20250824_160206_roc_20250824_160216.png	8/24/2025 9:32 PM	PNG File	25 KB	

Classification Report - random_forest_20250824_165415





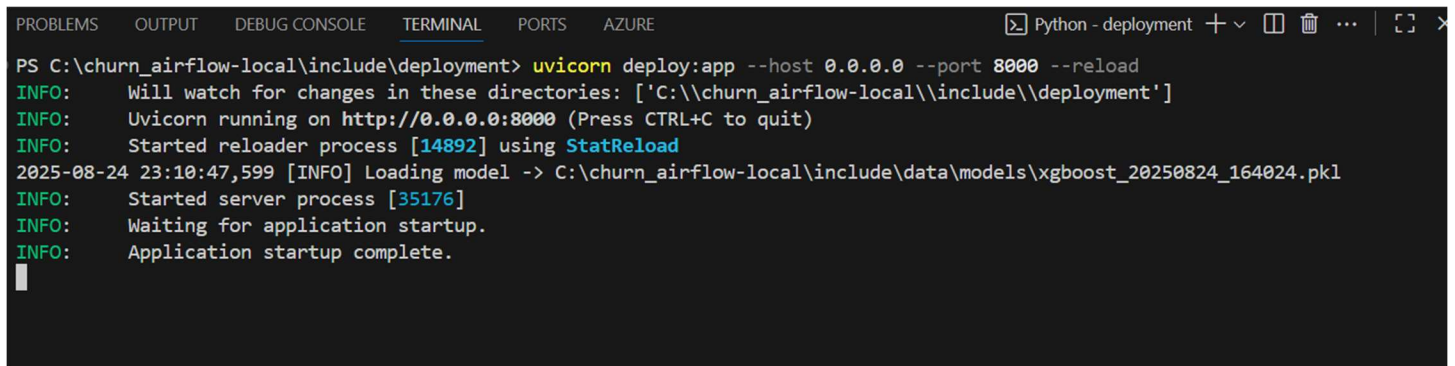
Random Forest Report

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1		precision	recall	f1-score	support									
2		0	0.903719	0.980652	0.940615									
3		1	0.966249	0.84131	0.899461									
4	accuracy		0.925333	0.925333	0.925333	0.925333								
5	macro avg		0.934984	0.910981	0.920038	3000								
6	weighted avg		0.928543	0.925333	0.924277	3000								
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

3.8 Deployment (deploy.py)

- Purpose: Deploy churn prediction model as REST API.
- Implementation:
 - FastAPI application with /predict endpoint
 - Loads model (model.pkl)
 - Accepts JSON input → returns churn prediction probability

Deploying Model



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE Python - deployment + - [ ] [ ] ... [ ] [ ] x
PS C:\churn_airflow-local\include\deployment> uvicorn deploy:app --host 0.0.0.0 --port 8000 --reload
INFO: Will watch for changes in these directories: ['C:\churn_airflow-local\include\deployment']
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [14892] using StatReload
2025-08-24 23:10:47,599 [INFO] Loading model -> C:\churn_airflow-local\include\data\models\xgboost_20250824_164024.pkl
INFO: Started server process [35176]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Request

```
{
  "features": {
    "age": 30,
    "monthly_charges": 80
  }
}
```

Response

```
{
  "model": "xgboost_20250824_164024.pkl",
  "prediction": 0,
  "probability": 0.33411720395088196,
  "missing_features": [
    "last_payment_date",
    "sign_up_date",
    "last_seen_at",
    "sessions_30d",
    "pages_viewed_30d",
    "time_on_site_sec_30d",
    "plan_premium",
    "plan_standard",
    "payment_status_delinquent",
    "payment_status_on_time",
    "device_type_desktop",
    "device_type_mobile",
    "device_type_tablet",
    "age_band_18-24",
    "age_band_25-34",
    "age_band_35-44",
    "age_band_45-54",
    "avg_monthly_spend_per_invoice",
    "engagement_score",
    "tenure_proxy",
    "avg_monthly_spend_per_invoice_scaled",
    "engagement_score_scaled",
    "tenure_proxy_scaled"
  ],
  "extra_features_ignored": []
}
```

3.9 Data Storage in Postgres

- Purpose: Store transformed features for analytics.
- Implementation:
 - Uses Airflow's PostgresHook to load latest transformed CSV into Postgres table churn_features.
 - Mode: replace ensures table updates daily.

	customer_id double precision	plan double precision	monthly_spend double precision	invoices double precision	last_payment_date double precision	payment_status double precision	sign_up_date double precision
1	-1.7314735534926655	-0.9522902528234478	-0.8184484978784055	1.240499109125429	0.39133510489320067	0.3125617392360236	-1.22915
2	-1.7303188528901363	-0.9522902528234478	-0.8011567885126515	-1.4404457028439166	1.0305668717141971	0.3125617392360236	1.41737
3	-1.729164152287607	-0.9522902528234478	-0.7170038029326493	-1.5269277935526053	0.9724548929122884	0.3125617392360236	1.54309
4	-1.7280094516850777	-0.9522902528234478	-0.5579200767677137	-0.48914270504834245	0.02716670440123904	0.3125617392360236	0.474429
5	-1.7268547510825483	1.2402429463095863	-0.054731334224276076	-0.9215531585917853	0.5734193051391815	0.3125617392360236	0.93018
6	-1.725700050480019	-0.9522902528234478	-0.6040313017430574	-0.48914270504834245	0.08527868320314781	0.3125617392360236	0.53414
7	-1.7245453498774899	-0.9522902528234478	-0.81671932694183	1.5864274719601832	-1.4798706125282617	0.3125617392360236	-1.6157
8	-1.7233906492749604	-0.9522902528234478	-0.7158510223082657	-1.180999430717851	0.9956996844330519	0.3125617392360236	1.15649
9	-1.7222359486724312	1.2402429463095863	-0.006314548000165241	-0.7485889771744081	0.3952092368133279	0.3125617392360236	0.71330
10	-1.7210812406699018	1.2402429463095863	-0.07029387265345453	1.0675349277080517	0.612160624340454	-3.199367915101322	-1.08142
11	-1.7199265474673726	1.2402429463095863	0.4916866817335463	-0.7485889771744081	0.9027205183499979	0.3125617392360236	0.7227
12	-1.7187718468648434	-0.9522902528234478	-0.8224832300637479	0.8945707462906747	-0.5423306878574669	0.3125617392360236	-0.89283
13	-1.717617146262314	1.2402429463095863	0.3107001237053224	-0.0567322515048996	-1.1195763439564272	0.3125617392360236	0.043819
14	-1.7164624456597848	-0.9522902528234478	-0.7273788285521017	0.4621602927472318	0.42232816025421865	0.3125617392360236	-0.459082
15	-1.7153077450572556	-0.9522902528234478	-0.4097877665344223	-0.22969643292227673	-0.7941492626657383	0.3125617392360236	0.26069
16	-1.7141530444547262	-0.9522902528234478	-0.4374545015196284	-0.48914270504834245	0.7400069777046533	0.3125617392360236	0.471286
17	-1.712998343852197	-0.9522902528234478	-1.053615745252658	0.9810528369993632	-2.1036058516687492	0.3125617392360236	-0.98084
18	-1.7118436432496675	-0.9522902528234478	-0.5740590055090841	-1.6998919749699823	1.3017561061231047	0.3125617392360236	1.68139

Total rows: 18000 Query complete 00:00:00.299 CRLF Ln 2, Col 1

4. Challenges & Solutions

1. File Path & Mounting Issues in Airflow Containers

- Challenge: Scripts tried to read/write from local relative paths (./data/...) which didn't exist inside the Airflow containers. Tasks failed with FileNotFoundError.
- Solution: Standardized all paths to container-safe directories (/usr/local/airflow/include/data/...) and mounted include/ into the container so both host and container see the same files.

2. Data Availability & Schema Inconsistency

- Challenge: CSV and JSON sources had missing values, different column orders, and inconsistent data types. The model training failed when unseen categorical values appeared.
- Solution: Added a validation step (validate.py) that:
 - Checked for missing values and duplicates.
 - Enforced column schema (numeric vs categorical).
 - Logged invalid records to a separate report.
 - Used imputation + encoding in preparation step to fix schema mismatches.

3. Large Data Handling in Pandas

- Challenge: Initial ingestion used Pandas to read entire large CSVs → memory usage spiked in Docker container.

- Solution: Switched to chunked reads (`pandas.read_csv(..., chunksize=10000)`) for ingestion and applied transformations in batches.

4. Airflow Task Failures & Debugging

- Challenge: Airflow tasks (`train_model`, `transform`) failed silently inside `BashOperator`. Debugging was hard because errors were buried in logs.
- Solution:
 - Added logging in each script (`logging.info`, `logging.error`).
 - Redirected logs to `/usr/local/airflow/dags/data/logs/` for easy inspection.
 - Used XComs to push metadata (like file paths, row counts) for better visibility in UI.

5. Model Versioning & Reproducibility

- Challenge: Multiple runs overwrote models in `/data/models/`, losing previous versions.
- Solution: Implemented model versioning by timestamping saved models (`model_2025-08-24.pkl`) and storing metadata (training date, algorithm, accuracy) in JSON logs.

6. Data Transformation Consistency Across Training & Inference

- Challenge: The model failed in deployment because transformation logic wasn't applied to incoming API requests.
- Solution: Extracted transformation logic into reusable functions and stored feature scaling parameters with the model (via `joblib`). The API applies the same transformations before prediction.

7. Postgres Integration

- Challenge: Loading transformed CSVs into Postgres sometimes failed due to datatype mismatches.
- Solution: Used Pandas `to_sql(..., dtype=...)` with explicit schema mapping (e.g., `VARCHAR`, `FLOAT`). Ensured Postgres table `churn_features` matched transformation outputs.

8. Pipeline Orchestration Dependencies

- Challenge: Sometimes downstream tasks (`train_model`) ran even when upstream (`transform`) produced empty/no file.
- Solution: Added validation checkpoints after each stage:
 - Check file existence + non-empty dataset before moving to next task.
 - Used Airflow `TriggerRule.ALL_SUCCESS` for critical dependencies.

9. Deployment & Port Exposure

- Challenge: FastAPI service (`deploy.py`) ran in container but wasn't reachable from host machine.
- Solution: Exposed port 8000 in `docker-compose.override.yml` and started API with:
 - `uvicorn deploy:app --host 0.0.0.0 --port 8000`

Verified at <http://localhost:8000/docs>.